Week 8 Deliverable

Preliminary FEA of Selected Rocker-Bogie Suspension Components

Conceptual Design of a Lunar Regolith Transport Rover

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Summer 2026

SOLIDWORKS Simulation linear-static study of the main rocker member, bogie subassembly, and wheel carrier. The work is a conceptual structural screening exercise, not flight qualification or final validation.

1 Week 8 Objective

The Week 8 objective was to use preliminary finite-element analysis (FEA) to determine whether the detailed rocker-bogie suspension geometry created in Week 6 appears structurally reasonable under the lunar loading developed in Week 5. Three manageable components were analyzed instead of attempting a nonlinear model of the entire six-wheel rover: the main rocker member, one bogie subassembly, and one wheel carrier. The studies report stress, displacement, factor of safety (FOS), load balance, and mesh sensitivity. The results are used to identify governing components and regions that need later refinement; they are not presented as flight qualification.

The analysis was completed in SOLIDWORKS Simulation using linear-elastic, small-displacement, static studies. The rocker-bogie architecture follows established planetary-rover mobility practice, while the current dimensions and joints are specific to this conceptual rover [6,7].

2 Relationship to Prior Deliverables

Week 3 established the rover mission as semi-autonomous transport of loose lunar regolith from a collection area to a habitat shielding area [1]. Week 4 established the first block-model geometry and an approximate overall rover envelope [2]. Week 5 established the mass, lunar wheel loading, and wheel-output torque requirements [3]. Week 6 replaced the block suspension with editable rocker members, bogie members, rectangular beam sockets, pivot housings, wheel mounts, and a complete mobility assembly [4]. Week 7 supplied the hand-analysis basis used to sanity-check the Week 8 results [5].

The Week 8 studies use the detailed Week 6 CAD geometry but intentionally simplify the joints. The selected FEA models therefore bridge the gap between the initial analytical sizing and a later full-rover structural model.

3 Components Selected for FEA

Table 1: Selected Week 8 FEA components and purpose.

Case	Component	Purpose
1	Main rocker member	Check global bending stress and displacement of the longest 800 mm rectangular hollow member and compare directly with beam theory.
2	Bogie subassembly	Check the 450 mm and 500 mm members with the central bogie joint under nominal, asymmetric, and rear-offset loading.
3	Wheel carrier	Check the combined carrier body under vertical hub load and continuous/peak wheel torque.

The optional Earth-handling load and a separate detailed bolt/pin contact study were not run. Those omissions are identified in Section 17 rather than being treated as completed analyses.

4 Geometry and Material Properties

The straight rocker and bogie members use a rectangular hollow section (RHS) with the 100 mm dimension oriented in the primary side-view bending plane. The Week 6 socket and pivot geometry was retained for the mixed bogie and solid carrier studies [4].

Table 2: Principal Week 6 geometry used in the Week 8 analysis.

Parameter	Value
RHS outside dimensions	100×50 mm
RHS wall thickness	5 mm
RHS inside dimensions	90×40 mm
Short bogie straight-member length	450 mm
Long bogie straight-member length	500 mm
Main rocker straight-member length	800 mm
Socket outside dimensions	120×70 mm
Socket opening	104×54 mm
Nominal socket wall thickness	8 mm
Typical beam engagement	125–140 mm
Primary retaining holes	approximately 13 mm for M12 bolts
Central smooth pivot pin	approximately 30 mm diameter

For the RHS, the cross-sectional area, second moment of area, and section modulus in the main bending direction are

$$A = b_o h_o - b_i h_i \tag{1}$$

$$= (50)(100) - (40)(90) = 1400 \text{ mm}^2, \tag{2}$$

$$I = \frac{b_o h_o^3 - b_i h_i^3}{12} \tag{3}$$

$$= \frac{(50)(100^3) - (40)(90^3)}{12} = 1.7367 \times 10^6 \text{ mm}^4, \tag{4}$$

$$S = \frac{I}{c} = \frac{1.7367 \times 10^6}{50} = 3.4733 \times 10^4 \text{ mm}^3. \tag{5}$$

The selected SOLIDWORKS material was **6061-T6 (SS)**. The values recorded from the material entry are consistent with standard handbook values for 6061-T6 aluminum [13,14].

Table 3: 6061-T6 properties used in every Week 8 study.

Property	Value	Units
Elastic modulus, E	6.90×10^{10}	N/m^2
Poisson ratio, ν	0.33	–
Shear modulus, G	2.60×10^{10}	N/m^2
Density, ρ	2700	kg/m^3
Ultimate tensile strength	3.10×10^{8}	N/m^2
Yield strength, σ_y	2.75×10^{8}	N/m^2

5 Lunar and Design Loading

The loaded rover mass includes the estimated 650 kg dry rover and 340 kg regolith payload:

$$m_{\text{loaded}} = 650 + 340 = 990 \text{ kg.} \tag{6}$$

Using lunar gravity $g_{\text{Moon}} = 1.62$ m/s^2,

$$W_{\text{Moon}} = mg = (990)(1.62) = 1603.8 \text{ N,} \tag{7}$$

$$N_{\text{wheel}} = \frac{1603.8}{6} = 267.3 \text{ N,} \tag{8}$$

$$F_{\text{wheel,design}} = 2.0(267.3) = 534.6 \text{ N.} \tag{9}$$

The 534.6 N design load already includes the complete loaded rover, including the regolith payload, and the preliminary factor of two. It must not be multiplied by two again. It is a simplified static design load that can also represent temporary load redistribution to fewer wheels; it is not an exact dynamic impact load.

The torque cases were 50 N m for continuous output and 100 N m for peak sensitivity. An Earth ground-handling sensitivity would give

$$N_{\text{wheel,Earth}} = \frac{(990)(9.81)}{6} = 1618.7 \text{ N,} \tag{10}$$

but this optional case was not solved and is not mixed with the lunar results.

6 FEA Modeling Approach

All studies used linear-elastic isotropic material behavior, static loading, and the SOLIDWORKS sparse linear solver. Loads were distributed over faces, bores, or beam joints rather than being applied at isolated vertices. SOLIDWORKS supports distributed forces and torques on appropriate geometric entities [8].

The element choice followed component thickness:

- Case 1 used beam elements with the RHS section properties.
- Case 2 used a mixed model: 5 mm thin shells for the RHS members and solid tetrahedral elements for the thick central joint.

- Case 3 used high-quality solid tetrahedral elements because the carrier is a thick, three-dimensional combined body.

Stress plots for solid and mixed models report von Mises stress. The Case 1 beam plot reports the SOLIDWORKS upper-bound axial-plus-bending stress, which is the appropriate beam quantity for comparison with the hand bending stress. Resultant displacement is reported in millimetres. The displayed deformation scale is exaggerated in all screenshots and is not the physical displacement scale.

For every case,

$$\mathrm{FOS} = \frac{\sigma_y}{\sigma_{\mathrm{reported}}}, \tag{11}$$

$$\mathrm{Utilization} = \frac{\sigma_{\mathrm{reported}}}{\sigma_y}(100\%). \tag{12}$$

7 Boundary Conditions and Contact Assumptions

Case 1

One beam endpoint was fixed and the 534.6 N design load was applied at the opposite endpoint. This cantilever idealization is intentionally simple and permits direct comparison with Euler-Bernoulli beam theory. It suppresses the rotation that a real pin/bushing may allow, so it is a global member screening model rather than a detailed pivot model.

Case 2

The central bogie pivot bore was restrained to establish static equilibrium. Loads were distributed at the two free beam ends. The 5 mm beam shells were bonded to the four corresponding internal socket faces of the solid central joint using local bonded interactions. SOLIDWORKS local interactions can connect selected solid, shell, and beam entities [9]. Bonding represents a rigid load path but does not reproduce through-bolt slip, bearing, preload, friction, hole tear-out, or contact separation.

Case 3

The carrier was combined into one solid body. The cylindrical surface of the upper steering/mounting bore was restrained against radial, axial, and circumferential motion. Vertical load was distributed over the central wheel-hub bore. Drive torque was distributed over the full wheel-side flange face about the hub axis. This avoids a nonphysical point force but still idealizes the bearing and steering interfaces.

The equilibrium checks were satisfactory. Case 1 returned 534.6 N and a 427.68 N m reaction moment. The Case 2 asymmetric and rear-offset studies returned approximately 801.9 N (reported by the user as near 800 N), equal to $267.3 + 534.6$ N. The added 15.39 N m offset torque changes reaction moment but adds no net force.

8 Mesh Strategy

High-quality elements and 16-point Jacobian checking were used for the mixed and solid studies. The SOLIDWORKS mesh-quality guidance identifies Jacobian ratio and aspect ratio as important checks for distorted tetrahedral elements [11]. The final solid and mixed meshes contained no distorted elements.

Table 4: Final mesh strategy by component.

Case	Element type	Final refinement	Local treatment
1	Beam	60 elements, 62 nodes	One-dimensional refinement along the 800 mm member.
2	Mixed shell/solid	7 mm maximum, 2 mm minimum	Shell thickness 5 mm; controls at socket transitions, bores, fillets, and bonded interfaces.
3	Solid tetrahedral	7 mm maximum, 2 mm minimum	Controls around hub bore, bolt holes, upper bore, fillets, and section transitions.

The Case 3 fine mesh had 535,457 nodes and 368,278 elements; 99.7% of elements had aspect ratio below 3, 0.115% had aspect ratio above 10, and none were distorted. The Case 2 fine asymmetric mesh had 1,332,730 nodes and 803,527 elements. The higher count results from the mixed shell/solid joint geometry and local interaction regions.

9 Case 1: Main Rocker Analysis

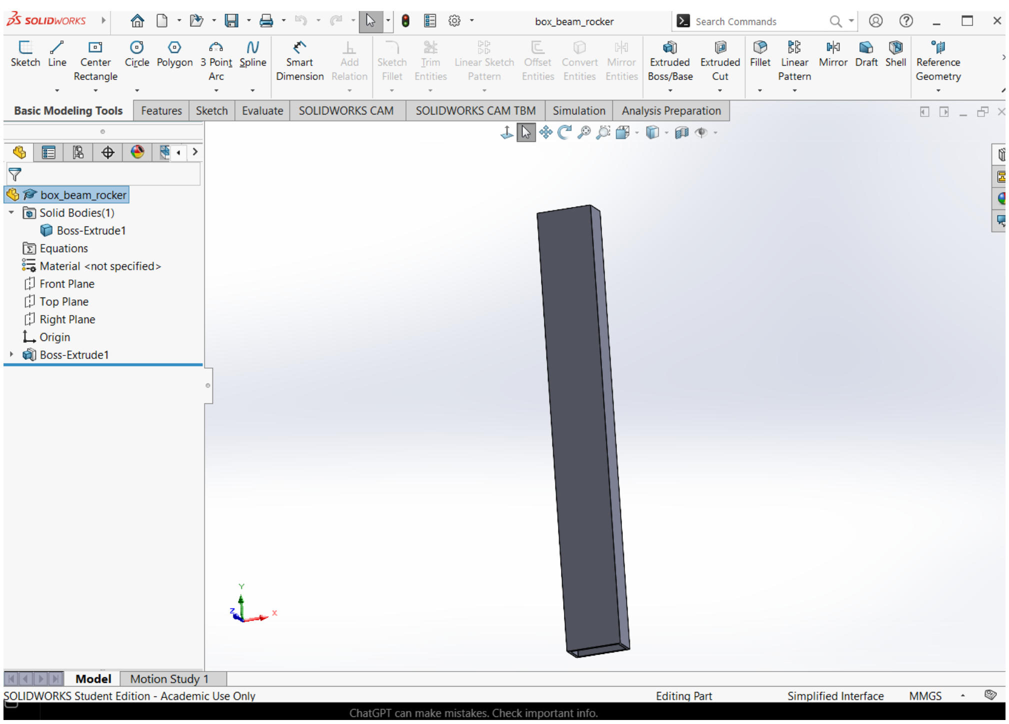


Figure 1: Case 1 CAD component: the 800 mm RHS rocker member. Source file: `fig01_case1_cad.png`.

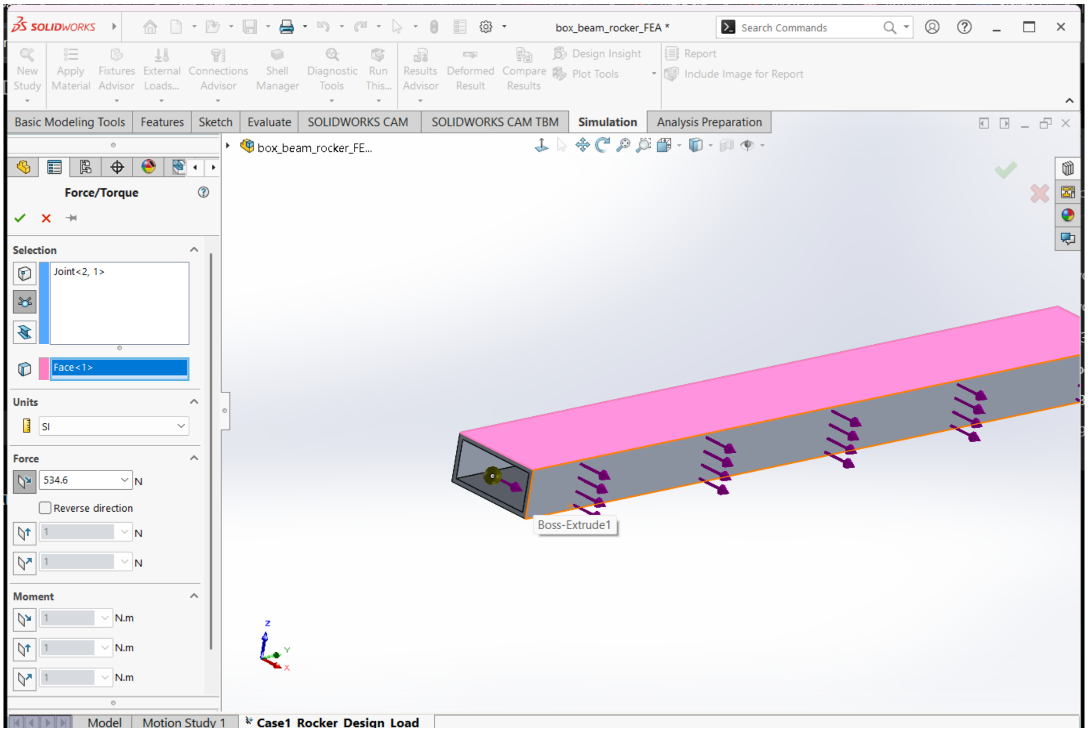


Figure 2: Case 1 fixture and distributed 534.6 N design load. Source file: `fig02_case1_loads.png`.

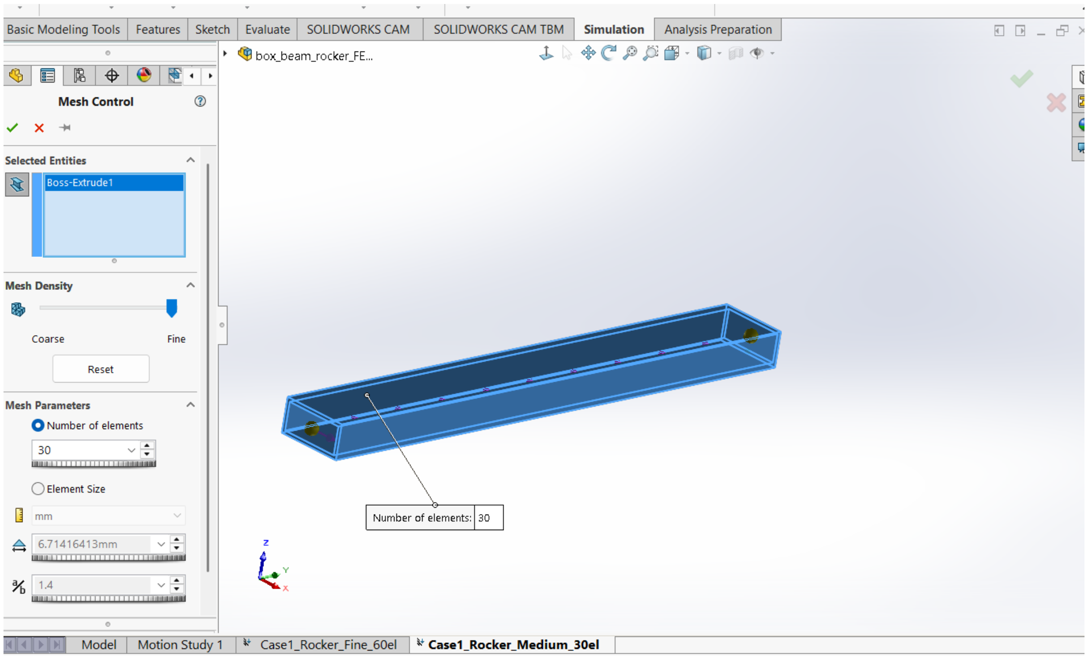


Figure 3: Case 1 beam mesh. The convergence sequence used 10, 30, and 60 beam elements. Source file: `fig03_case1_mesh.png`.

The design-load beam study produced an upper-bound axial-plus-bending stress of 12.31 MPa and a maximum displacement of 0.7799 mm. The maximum displacement occurs at the loaded end. The stress distribution rises linearly toward the fixed end, as expected for a cantilever under an end force.

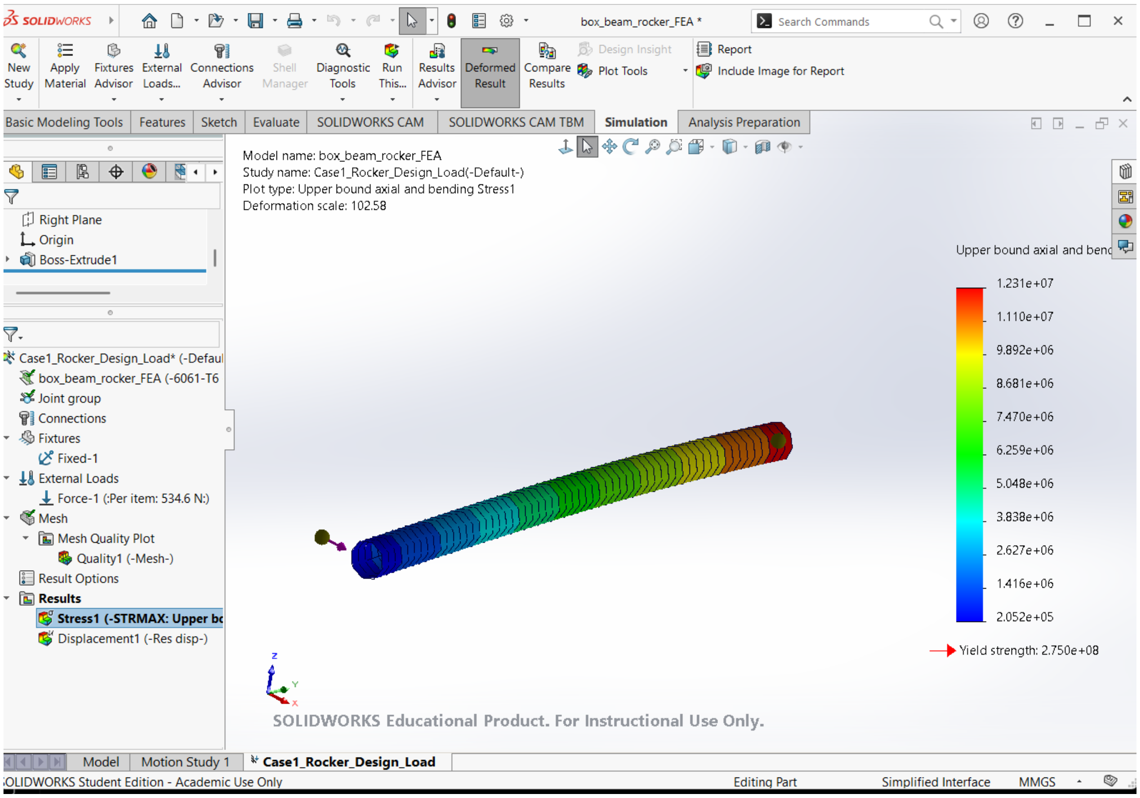


Figure 4: Case 1 upper-bound axial and bending stress, with a maximum of 12.31 MPa. Source file: `fig04_case1_stress.png`.

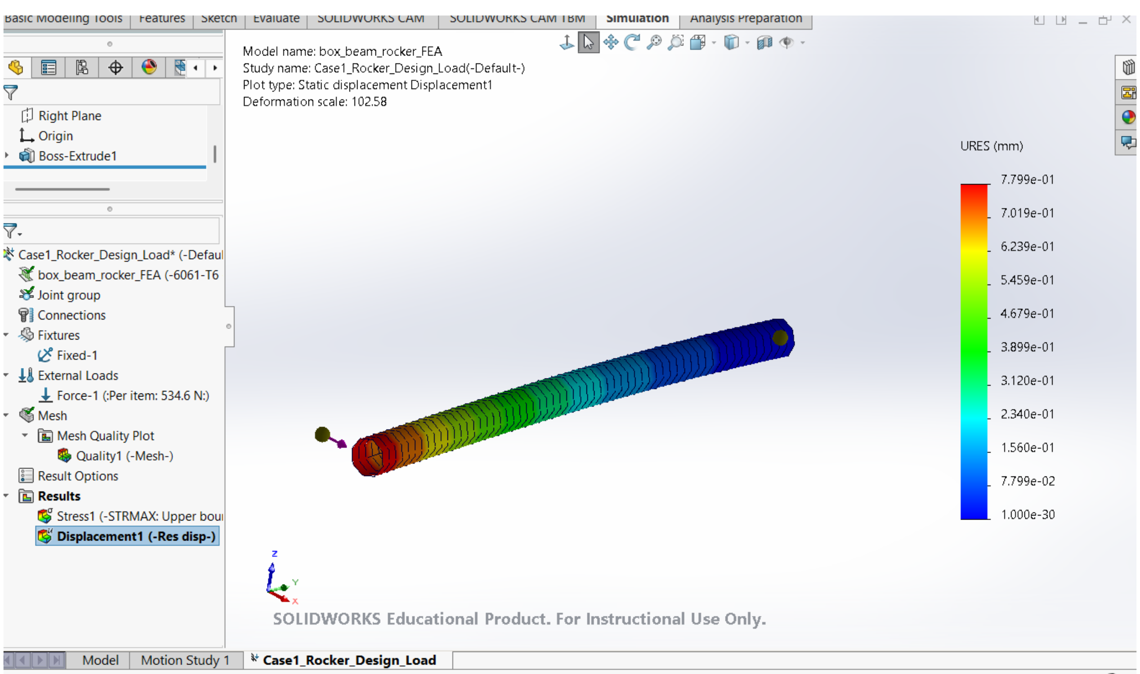


Figure 5: Case 1 resultant displacement, with a maximum of 0.7799 mm at the loaded end. Source file: `fig05_case1_displacement.png`.

$$\mathrm{FOS}_1 = \frac{275}{12.31} = 22.34, \tag{13}$$

$$\mathrm{Utilization}_1 = \frac{12.31}{275}(100) = 4.48\%. \tag{14}$$

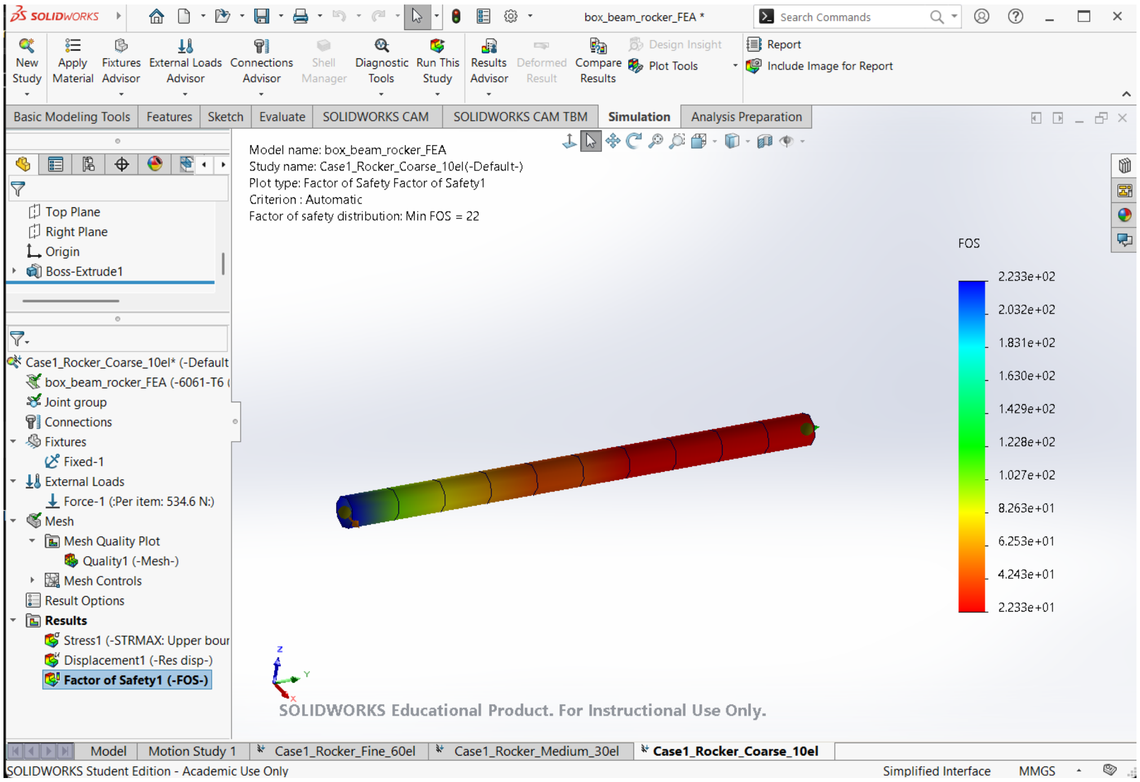


Figure 6: Case 1 factor-of-safety plot. SOLIDWORKS reported a minimum FOS of approximately 22.3. Source file: `fig06_case1_fos.png`.

Case 1 result: PASS for the idealized static design load. The 12.31 MPa beam stress is only 4.48% of the assumed 275 MPa yield strength. This pass applies to the global RHS member model, not to local bolt holes, socket bearing, pivot contact, buckling, fatigue, or impact.

10 Case 1 Hand-Calculation Comparison

For the 800 mm cantilever and 534.6 N end load,

$$M_{\max} = FL = (534.6)(800) = 427{,}680\,\mathrm{N\,mm} = 427.68\,\mathrm{N\,m}. \tag{15}$$

The elastic bending stress is

$$\sigma_{\max} = \frac{M_{\max}c}{I} = \frac{(427{,}680)(50)}{1.7367\times10^6} = 12.313\,\mathrm{MPa}. \tag{16}$$

The Euler-Bernoulli tip displacement is

$$\delta_{EB} = \frac{FL^3}{3EI} \tag{17}$$

$$= \frac{(534.6)(800^3)}{3(69{,}000)(1.7367\times10^6)} = 0.7614\,\mathrm{mm}. \tag{18}$$

The FEA stress agrees with the hand result to the reported precision. The displacement difference is

$$\frac{|0.7799 - 0.7614|}{0.7614}(100) = 2.43\%. \tag{19}$$

An approximate rectangular-section shear term using $k = 5/6$, $G = 26{,}000\ \mathrm{N/mm^2}$, and $A = 1400\ \mathrm{mm^2}$ is

$$\delta_s = \frac{FL}{kGA} = 0.01410\,\mathrm{mm}, \tag{20}$$

which gives $\delta_{EB} + \delta_s = 0.7755\,\mathrm{mm}$, only 0.57% below the FEA result. The remaining difference is consistent with beam formulation, boundary-condition implementation, and section shear effects.

Quantity	Hand calculation	FEA	Difference
Stress	12.313 MPa	12.31 MPa	0.03% (rounding)
Displacement, Euler-Bernoulli	0.7614 mm	0.7799 mm	2.43%
Displacement, with shear estimate	0.7755 mm	0.7799 mm	0.57%

Figure 7: Case 1 hand calculation compared with the beam-element FEA result.

11 Case 2: Bogie Analysis

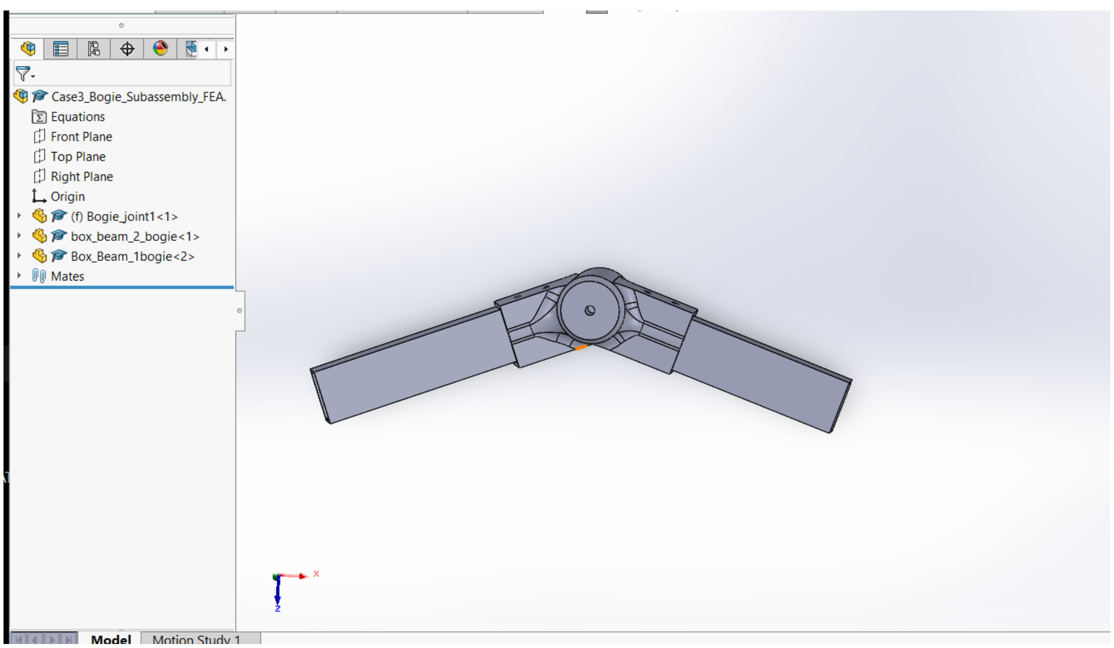


Figure 8: Case 2 bogie subassembly containing the central joint and the 450 mm and 500 mm RHS members. Source file: `fig07_case2_cad.png`.

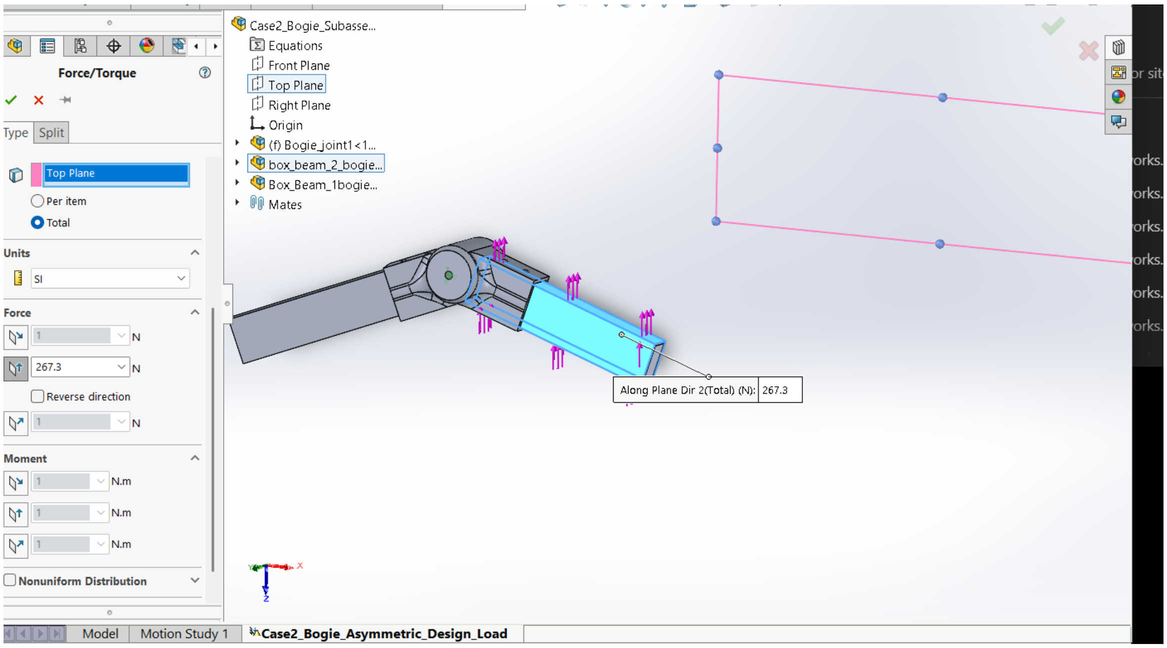


Figure 9: Case 2 load setup. The final asymmetric study used 267.3 N at one end and 534.6 N at the other, with the central pivot restrained. Source file: `fig08_case2_loads.png`.

The primary design study was asymmetric: 267.3 N at one wheel location and 534.6 N at the other. The summed reaction was approximately 801.9 N. The fine study produced a raw maximum von Mises stress of 21.85 MPa, maximum displacement of 0.1890 mm, and computed FOS of 12.58. The highest stresses occur near the central pivot/socket transition and bonded joint interfaces. Because these local peaks were mesh sensitive, the mesh convergence decision used a probe on a smooth beam wall away from the fixed/bonded discontinuities.

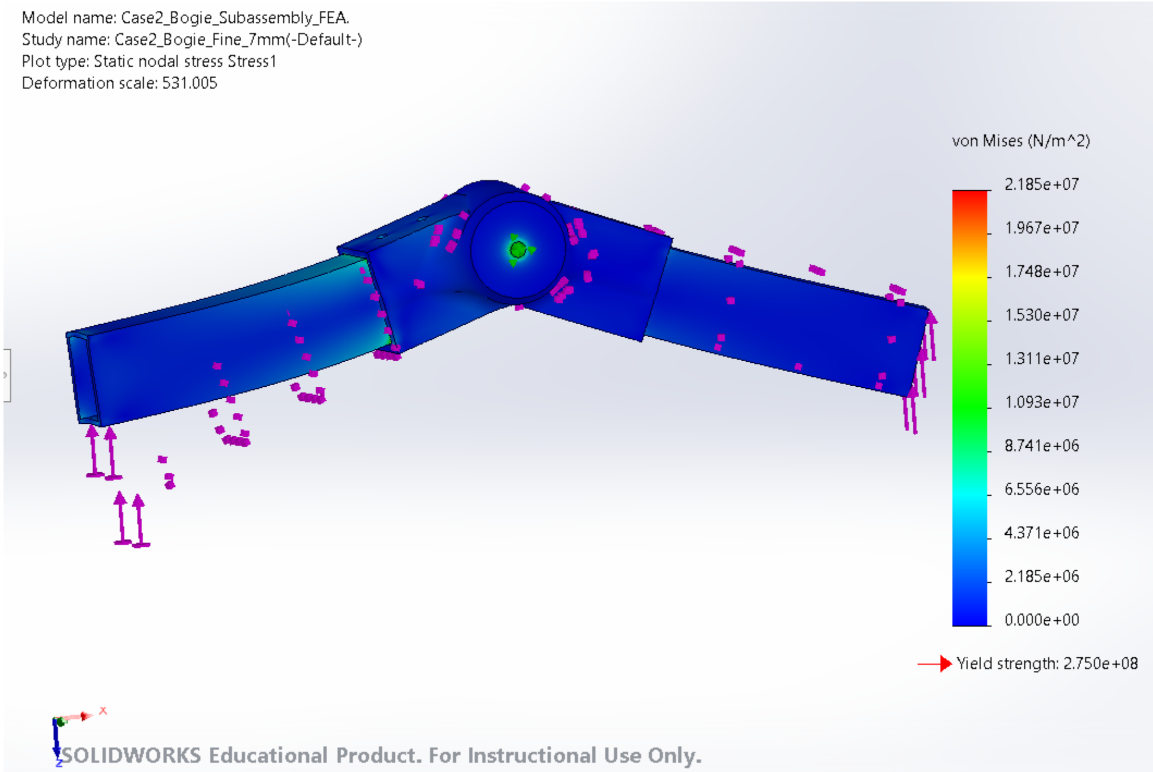


Figure 10: Fine-mesh Case 2 asymmetric von Mises stress plot; raw maximum 21.85 MPa. Source file: `fig09_case2_stress.png`.

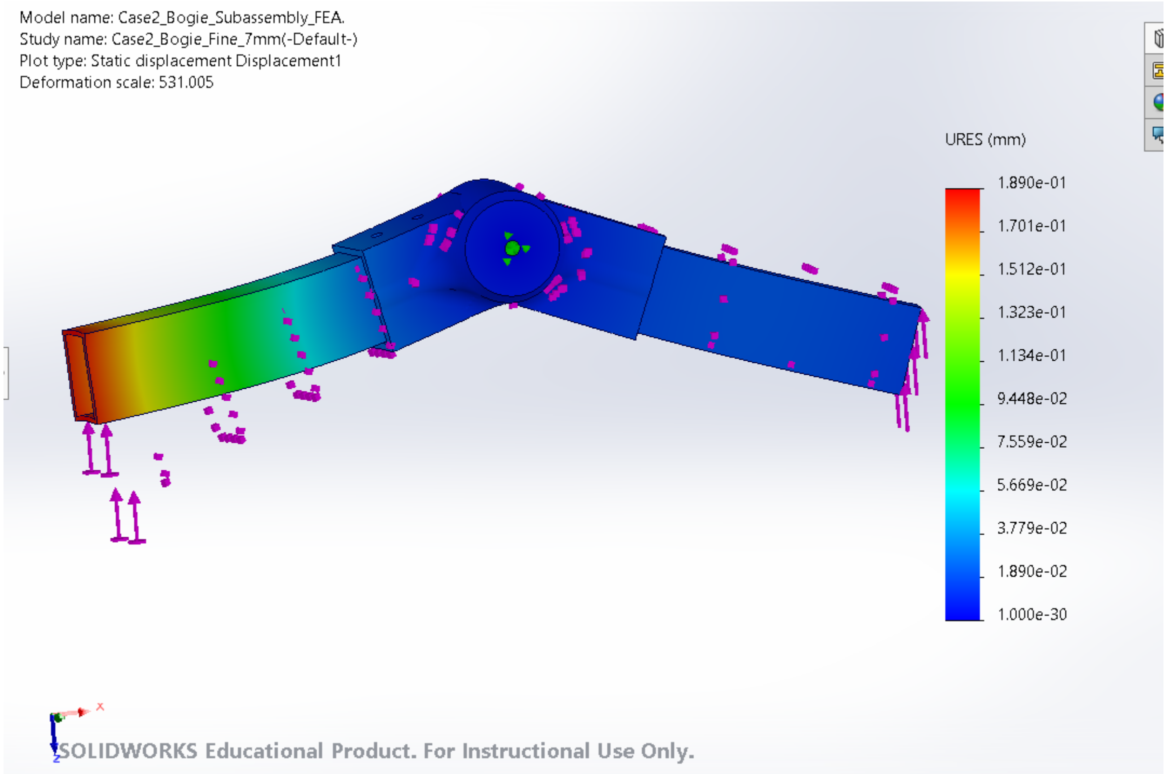


Figure 11: Fine-mesh Case 2 asymmetric resultant displacement; maximum 0.1890 mm. Source file: `fig10_case2_displacement.png`.

Selected fine-mesh probe values recorded around the bogie joint were 8.898 MPa, 7.183 MPa, and 5.676 MPa. The representative smooth-wall probe was 1.079 MPa at approximately $(X, Y, Z) = (192, 2350, 1040)$ mm (node 854058). The corresponding medium-mesh probe was 1.040 MPa at approximately $(194, 2350, 1050)$ mm (node 752741).

$$\mathrm{FOS}_{2,\mathrm{fine}} = \frac{275}{21.85} = 12.58, \tag{21}$$

$$\mathrm{Utilization}_{2,\mathrm{fine}} = \frac{21.85}{275}(100) = 7.95\%. \tag{22}$$

The medium mesh produced the highest raw local peak, 27.18 MPa. Treating that as the conservative screened value gives

$$\mathrm{FOS}_{2,\mathrm{conservative}} = \frac{275}{27.18} = 10.12, \qquad \mathrm{Utilization} = 9.88\%. \tag{23}$$

The nominal symmetric bogie study applied 267.3 N at each end. It produced 12.93 MPa maximum stress, 0.06646 mm maximum displacement, and FOS 21.26. The asymmetric study is therefore the more demanding of the two vertical-only bogie cases.

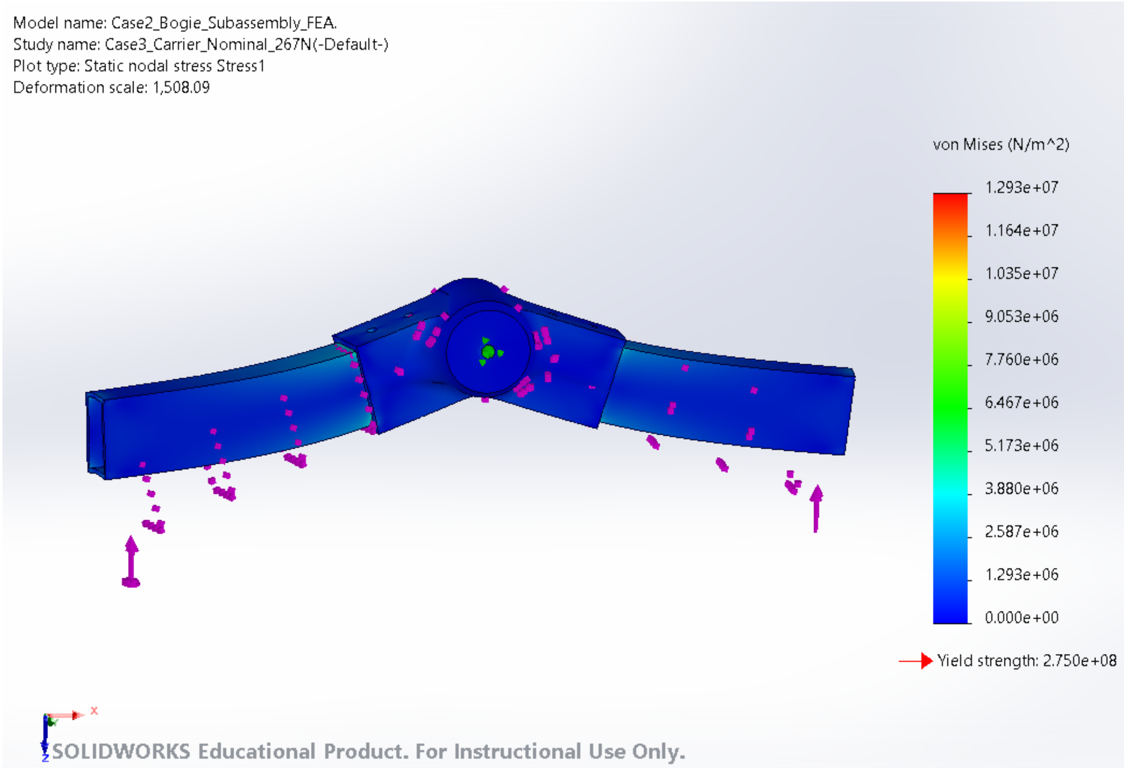
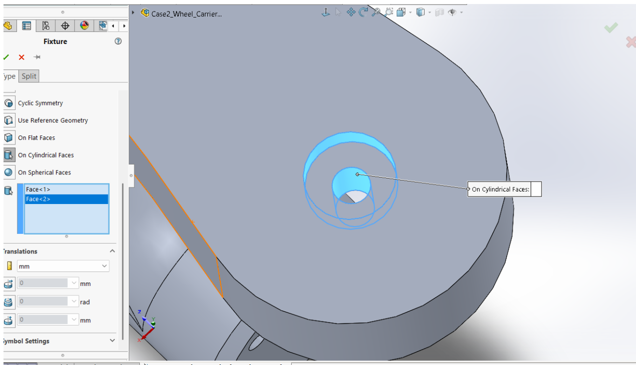


Figure 12: Nominal symmetric bogie study with 267.3 N at each end; maximum stress 12.93 MPa. Source file: `fig17_case2_nominal_stress.png`.

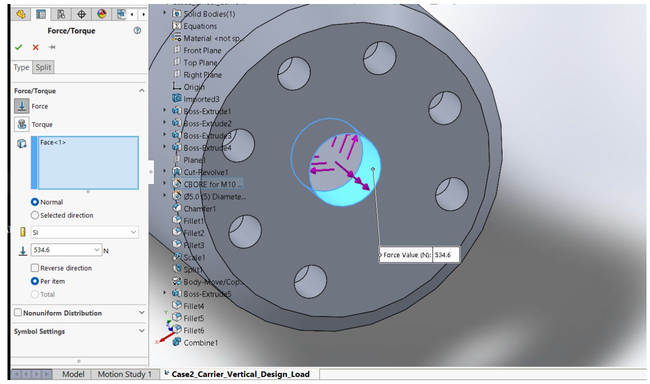
Case 2 result: PASS for the idealized bonded static studies. Even the conservative raw mesh peak of 27.18 MPa gives FOS 10.12. The result does not validate the pivot pin, bushings, retaining bolts, shell-to-solid load transfer, or fatigue life.

12 Case 3: Wheel-Mount or Pivot Analysis

The wheel carrier was combined into a single solid body before meshing. The upper bore was restrained and vertical wheel load was distributed through the central hub bore. The solid model used 6061-T6 and high-quality blended curvature-based tetrahedral elements.



(a) Upper-bore cylindrical fixture.



(b) Distributed hub-bore load.

Figure 13: Case 3 wheel-carrier setup. Source files: `fig11a_case3_fixture.png` and `fig11b_case3_load.png`.

Table 5: Case 3 wheel-carrier load-case results using the fine mesh.

Load case	Vertical load (N)	Torque (N m)	Stress (MPa)	Displacement (mm)	FOS
Nominal lunar	267.3	0	2.121	0.01920	129.66
Design vertical	534.6	0	4.242	0.03839	64.83
Continuous torque	534.6	50	6.170	0.05341	44.57
Peak torque sensitivity	534.6	100	9.577	0.08265	28.71

The nominal case is exactly one-half of the design-vertical response to the displayed precision, which is consistent with a linear-static model. Relative to the design-vertical case, the 50 N m case increased stress by 45.5% and displacement by 39.1%. The 100 N m case increased stress by 125.8% and displacement by 115.3%. The torque cases move the critical stress toward the hub bore and adjacent transition, which is physically consistent with combined bending and torsion.

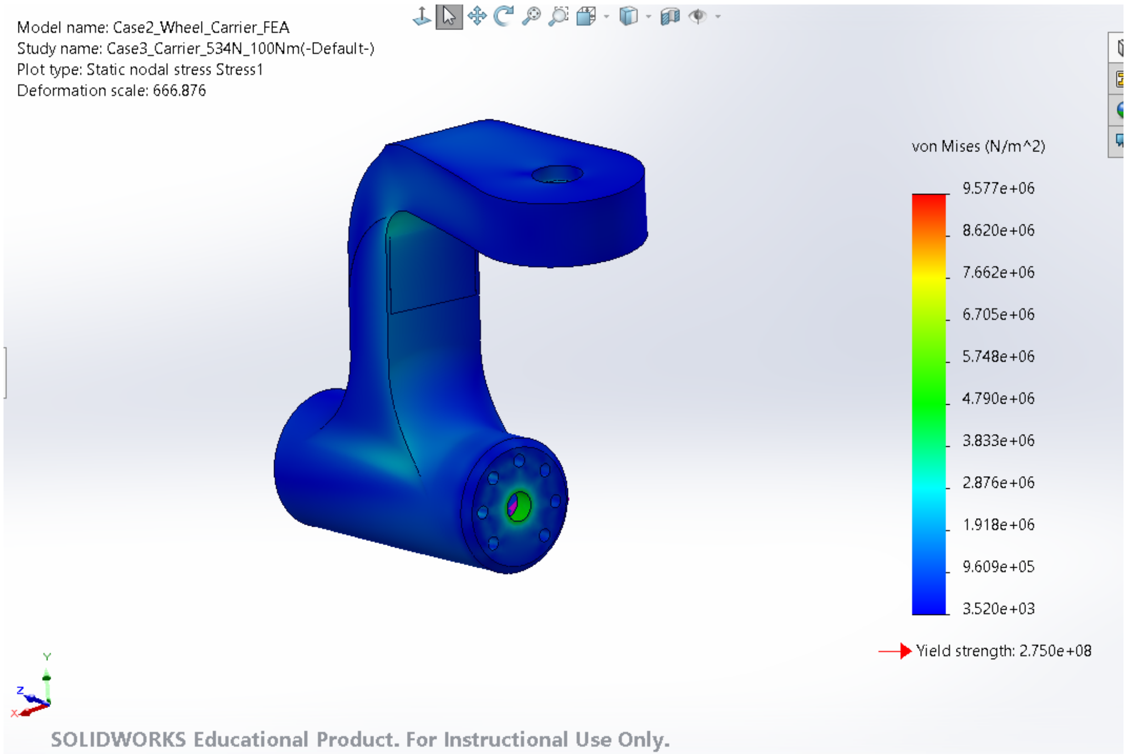


Figure 14: Case 3 peak-torque sensitivity stress plot for 534.6 N + 100 N m; maximum 9.577 MPa near the hub-bore load path. Source file: `fig12_case3_peak_stress.png`.

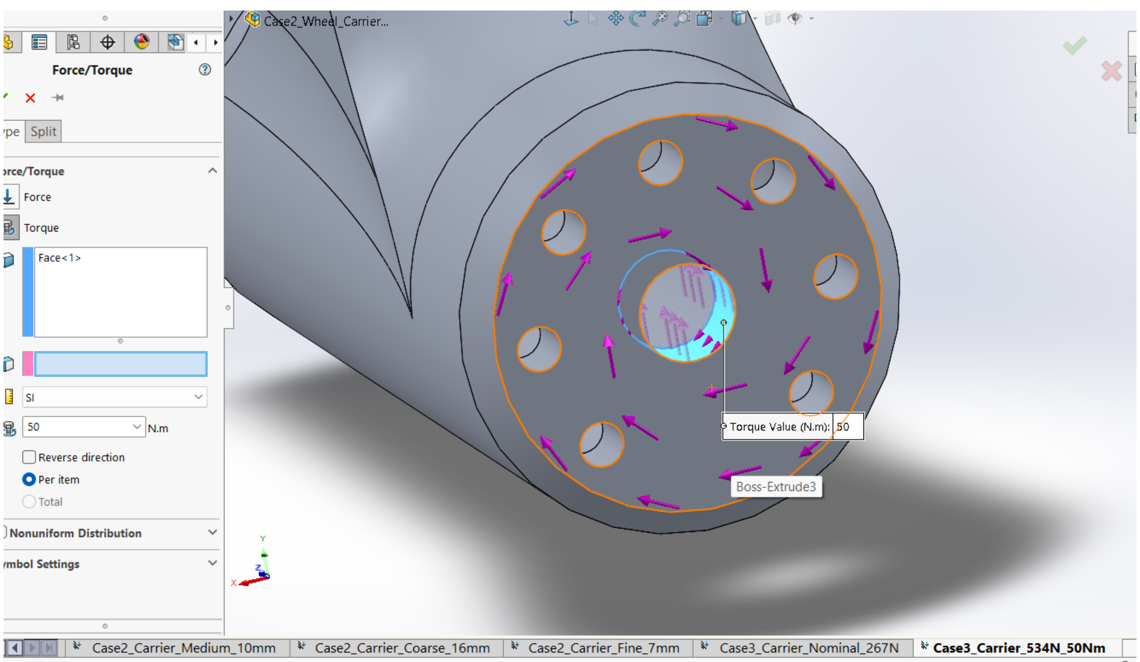


Figure 15: Torque distributed over the full wheel-side flange face about the hub axis. Source file: `fig18_case3_torque_setup.png`.

For the governing 100 N m carrier study,

$$\mathrm{FOS}_{3,100} = \frac{275}{9.577} = 28.71, \tag{24}$$

$$\mathrm{Utilization}_{3,100} = \frac{9.577}{275}(100) = 3.48\%. \tag{25}$$

Case 3 result: PASS for all four idealized static load combinations. The 100 N m sensitivity is the governing carrier case but remains below 4% yield utilization. Bearing contact, steering loads, motor-mount fasteners, wheel impacts, fatigue, and local bolt-hole bearing remain outside this model.

13 Rear-Wheel Offset Sensitivity

The completed rover assembly was measured between the middle and rear wheel centers. The recorded offsets were $dX = 1254.82$ mm, $dY = 28.78$ mm, and $dZ = 257.09$ mm, with a three-dimensional center distance of 1281.21 mm. The lateral offset relevant to bogie torsion is $e = dY = 28.78\,\text{mm} = 0.02878$ m.

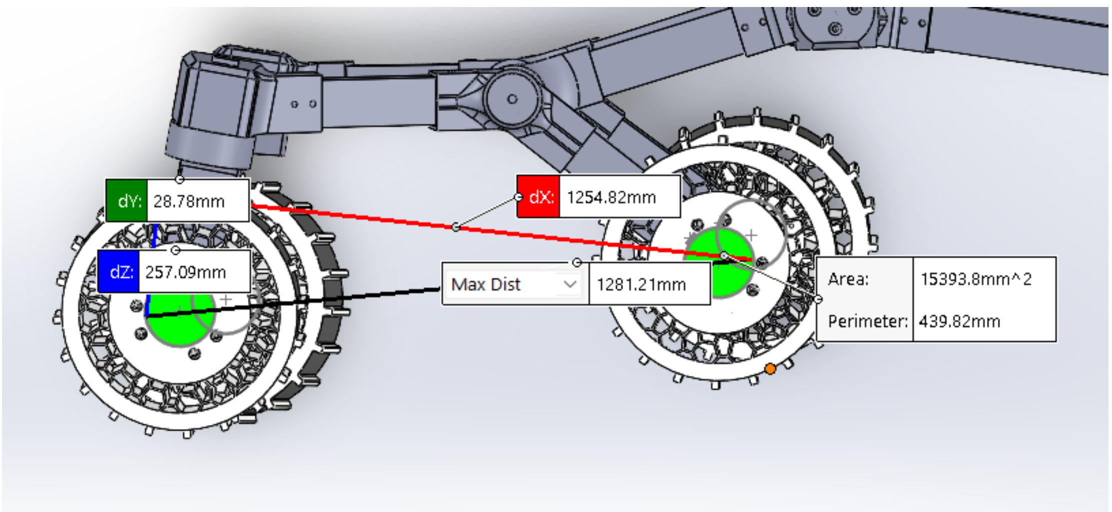


Figure 16: CAD measurement between rear and middle wheel centers. The lateral offset used for torsion is $dY = 28.78$ mm. Source file: `fig15_rear_offset_measurement.png`.

Using the design wheel load,

$$M_{\text{offset}} = F_{\text{design}}e = (534.6)(0.02878) = 15.39\,\text{N m}. \tag{26}$$

The fine asymmetric Case 2 study was copied, the original 267.3 N and 534.6 N vertical loads were retained, and 15.39 N m was applied about the longitudinal axis of the rear member.

Table 6: Effect of the measured rear-wheel lateral offset.

Result	Asymmetric vertical only	With 15.39 N m offset	Increase
Maximum stress	21.85 MPa	39.42 MPa	80.4%
Maximum displacement	0.1890 mm	0.2819 mm	49.2%
Minimum FOS	12.58	6.98	–
Utilization	7.95%	14.33%	–

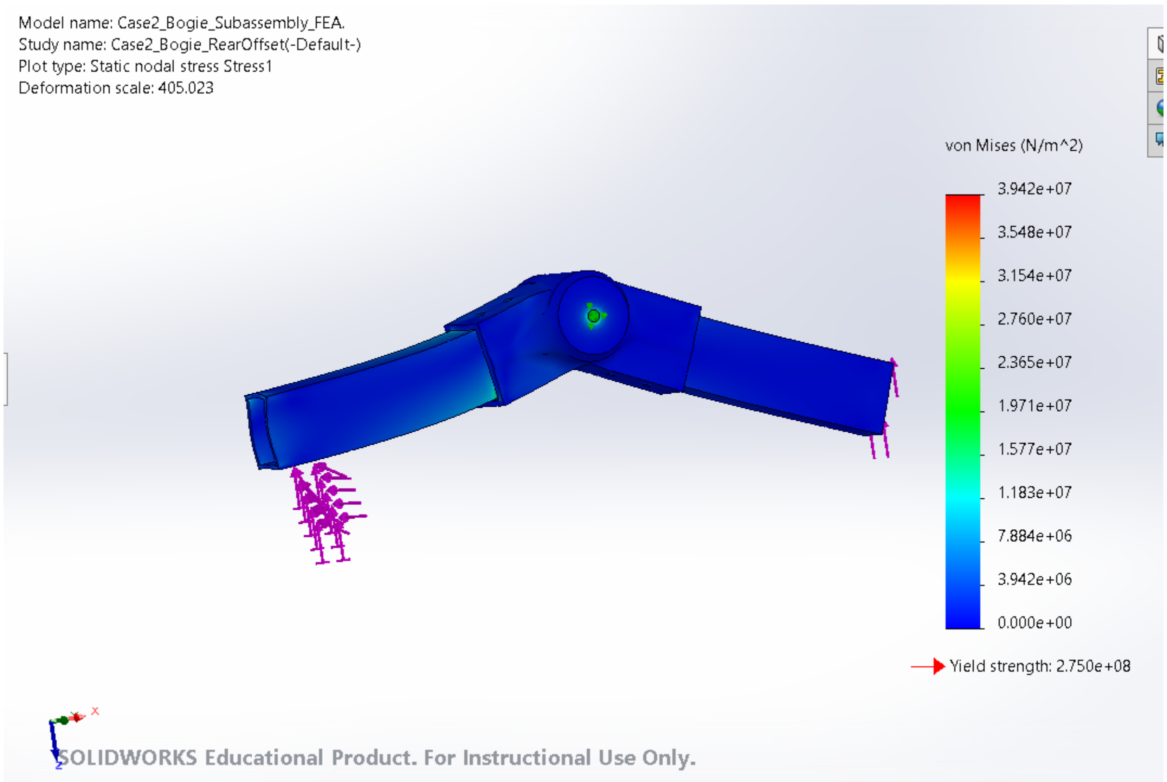


Figure 17: Rear-offset sensitivity stress plot; maximum 39.42 MPa. Source file: `fig16_rear_offset_stress.png`.

The rear-offset study is the governing Week 8 result. It remains below yield, but the 80.4% stress increase demonstrates that even a 28.78 mm lateral wheel offset materially raises bogie torsional demand. The equilibrium check remained near 801.9 N because a pure applied torque adds moment but no net force.

14 Mesh-Convergence Study

Three mesh levels were run where practical. Case 1 used 10, 30, and 60 beam elements. Case 2 used mixed shell/solid meshes with 16/5 mm, 10/2.5 mm, and 7/2 mm maximum/minimum element sizes. Case 3 used solid meshes with 16/5 mm, 10/3 mm, and 7/2 mm settings.

Table 7: Mesh-convergence record.

Case	Mesh	Nodes	Elements	Reported stress (MPa)	Displacement (mm)	FOS
1 rocker	Coarse	12	10	12.31	0.7799	22.34
1 rocker	Medium	32	30	12.31	0.7799	22.34
1 rocker	Fine	62	60	12.31	0.7799	22.34
2 bogie asymmetric	Coarse	1,271,480	762,971	24.08	0.1886	11.42
2 bogie asymmetric	Medium	1,118,501	642,429	27.18	0.1887	10.12
2 bogie asymmetric	Fine	1,332,730	803,527	21.85	0.1890	12.58
3 carrier design	Coarse	271,564	180,631	4.190	0.03839	65.63
3 carrier design	Medium	305,914	204,901	4.264	0.03839	64.49
3 carrier design	Fine	535,457	368,278	4.242	0.03839	64.83

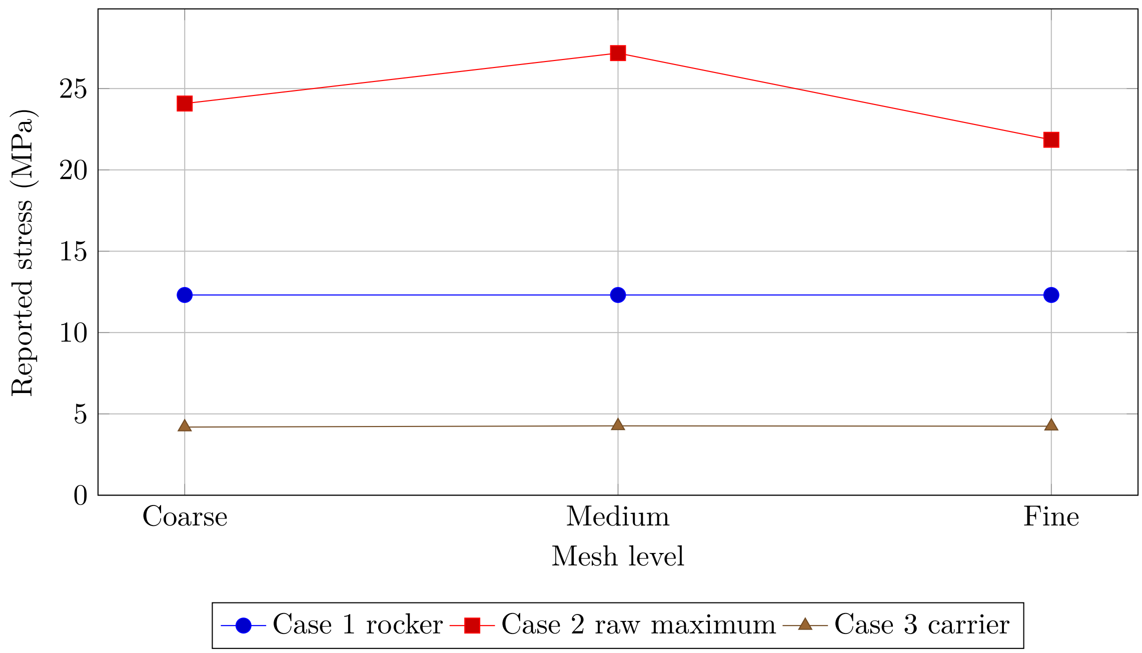


Figure 18: Stress convergence across the three component studies. The Case 2 raw peak is nonmonotonic because it occurs near a bonded/fixture discontinuity; a smooth-wall probe was used for the acceptance check.

For Case 2, the representative smooth-wall stress changed from 1.040 MPa on the medium mesh to 1.079 MPa on the fine mesh:

$$\Delta\sigma_{\text{rep}} = \frac{|1.079 - 1.040|}{1.040}(100) = 3.75\%. \tag{27}$$

The maximum displacement changed by

$$\Delta\delta = \frac{|0.1890 - 0.1887|}{0.1887}(100) = 0.16\%. \tag{28}$$

For Case 3, the medium-to-fine stress change was

$$\frac{|4.242 - 4.264|}{4.264}(100) = 0.52\%, \tag{29}$$

and the displayed displacement change was 0%. Case 1 was unchanged to the displayed precision. The final meshes therefore meet the conceptual-stage target of less than approximately 5–10% change for representative nonsingular stress and displacement.

15 Comparison of FEA Cases

Table 8: Comparison of all completed Week 8 load cases.

Component	Load case	Stress (MPa)	Displ. (mm)	FOS	Utilization	Result
Rocker	534.6 N	12.31	0.7799	22.34	4.48%	Pass
Bogie	267.3 N each	12.93	0.06646	21.26	4.70%	Pass
Bogie	267.3 + 534.6 N	21.85	0.1890	12.58	7.95%	Pass
Bogie	Conservative raw mesh peak	27.18	0.1887	10.12	9.88%	Pass
Bogie	Asymmetric + 15.39 N m offset	39.42	0.2819	6.98	14.33%	Pass
Carrier	267.3 N	2.121	0.01920	129.66	0.77%	Pass
Carrier	534.6 N	4.242	0.03839	64.83	1.54%	Pass
Carrier	534.6 N + 50 N m	6.170	0.05341	44.57	2.24%	Pass
Carrier	534.6 N + 100 N m	9.577	0.08265	28.71	3.48%	Pass

The rear-offset bogie case governs stress and FOS. The long rocker governs displacement because it is modeled as an 800 mm cantilever. The solid carrier remains the least utilized component even under peak torque. Large FOS values should be interpreted as evidence that the idealized static loads are low relative to the provisional material yield strength, not as proof that the joints are fully validated.

16 Design Interpretation

The three components appear structurally reasonable for the stated preliminary lunar static loads. Case 1 shows that the $100 \times 50 \times 5$ mm RHS has low global bending stress, and the hand/FEA agreement confirms that the beam model is internally consistent. Case 2 shows that the central joint and socket transitions are more critical than the straight beams. The rear offset is important: a 28.78 mm lateral offset increased the modeled bogie stress by 80.4% and displacement by 49.2%. Case 3 shows that the combined carrier body has low stress under both vertical load and drive torque; nevertheless, the hub bore and inner fillet remain the appropriate local refinement areas.

The current geometry is likely conservative for these idealized static loads, but structural mass should not be removed solely from this study. The very high FOS values are partly caused by low lunar gravity, the preliminary load factor, rigid bonded connections, and omitted local contact/fastener effects. Any later weight reduction must retain adequate margins for wheel impacts, obstacle climbing, launch/handling, thermal gradients, fatigue, and joint bearing.

17 Limitations

- The studies are linear static with small displacements. Dynamic wheel impact, obstacle climbing, vibration, and nonlinear material behavior are excluded.
- The 275 MPa 6061-T6 yield strength is provisional. Welded or heat-affected regions may have substantially lower strength.
- The Case 1 fixed-end beam is a cantilever idealization and does not reproduce detailed pivot/bushing compliance.
- The Case 2 beam-to-socket interfaces are bonded. Bolt slip, preload, bearing stress, friction, separation, tear-out, and beam-wall crushing are not resolved.
- The Case 2 central pivot restraint can create mesh-sensitive local peaks. Representative smooth-wall probes were therefore used for convergence.
- Shell thickness was assigned as 5 mm at the middle surface. Manufacturing variation, corner radii, welds, and local wall thinning are not included.
- The Case 3 upper-bore fixture and hub-bore load simplify the actual bearings, steering shaft, motor, gearbox, and fasteners.
- The rear-wheel offset is represented by an equivalent 15.39 N m torque derived from the measured 28.78 mm lateral offset. It is not a full wheel-ground contact model.
- The displayed deformed shapes are highly magnified and must not be interpreted as physical scale.
- Local stress at sharp corners, perfectly fixed edges, and bonded transitions may be singular or mesh dependent.
- Buckling, fatigue, crack growth, wear, dust, vacuum, thermal cycling, radiation, lubrication, and corrosion are excluded.
- The optional Earth ground-handling sensitivity was calculated but not solved.
- The results do not qualify the rover for manufacturing, launch, lunar operation, or human safety. They are preliminary conceptual screening results only.

18 Engineering Summary

Week 8 completed preliminary FEA of the main rocker, bogie subassembly, and wheel carrier using the detailed Week 6 CAD. The lunar loaded weight is 1603.8 N, the nominal wheel load is 267.3 N, and the factor-two design wheel load is 534.6 N. All studies used provisional 6061-T6 aluminum with 275 MPa yield strength.

The main rocker produced 12.31 MPa stress, 0.7799 mm displacement, and FOS 22.34. Its hand-calculated stress was 12.313 MPa and Euler-Bernoulli displacement was 0.7614 mm, giving 2.43% displacement difference. The asymmetric bogie produced 21.85 MPa fine-mesh stress and 0.1890 mm displacement; a representative medium-to-fine stress probe converged within 3.75%. The measured 28.78 mm rear-wheel lateral offset raised the bogie result to 39.42 MPa, 0.2819 mm, and FOS 6.98, making it the governing study. The wheel carrier produced 4.242 MPa under design vertical load, 6.170 MPa with 50 N m, and 9.577 MPa with 100 N m; the peak-torque FOS was 28.71.

All completed idealized cases pass yielding with the provisional material. The strongest design conclusion is not simply that FOS exceeds one, but that rear-wheel offset and central joint transitions deserve the most attention in later joint, contact, fastener, fatigue, and dynamic studies.

19 References

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